SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I KARNESH.V.K / 192524296 (Name / Reg. No.)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of data centers in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

cloud computing:

cloud computing is the delivery of computing services such as servers, storage, databases and software over the internet.

4 - characteristics:

→ On-demand self-service:

Users can access computing resources whenever required without contacting the service provider.

→ Broad network access:

cloud services are accessible through the Internet using laptops, mobiles & other devices.

→ Resource pooling:

Computing resources are shared among multiple users to improve utilization

→ Rapid elasticity:

Resources can be increased or decreased quickly according to user demand.

Difference: Traditional computing requires organizations to maintain their own hardware, while cloud computing provides resources over the internet.

2) Evolution of cloud computing:

cloud computing evolved through several stages.

→ Mainframe computing:

Large centralized computers served multiple users.

→ Personal computers:

Computing became available to individual users.

→ client-server computing:

clients accessed data and services from centralized servers.

→ virtualization:

Multiple virtual machines could run on a single physical server.

→ Internet evolution:

Internet technologies allowed users to access applications and data online

→ Modern cloud-computing:

Computing resources became available as scalable services over the internet

Eg:- Amazon web sevices (AWS), Microsoft Azure & google cloud.

3) Server virtualization is a technology that allows multiple machines to run on a single physical server using software called a hypervisor.

Role of virtualization:

→ Improves hardware utilization

→ Reduces hardware & maintenance costs

→ Allow multiple operating system to run on one server

→ Supports flexible allocation of computing resources.

Virtualization	Cloud Computing
Creates virtual machines from physical hardware	Delivers computing resources as services
Improves hardware utilization	Provides on-demand access to resources
Can be used within a private organization	Commonly provides internet-accessible services

Eg:- VMware

Eg:- AWS

Conclusion: Virtualization is an enabling technology, whereas cloud computing uses technologies such as virtualization to deliver services to users.

4) A data center is a facility that contains servers, storage devices, networking equipment & supporting infrastructure.

Roles of data centers:

→ Data Storage:

Stores large amounts of user & application data

→ Data processing:

Process information & runs cloud applications

→ Network Connectivity:

Connects servers & delivers cloud services over the internet.

→ Security:

Supports data protection & physical security

→ Reliability : uses backup systems & redundant equipment to reduce service interruptions

→ scalability : Allows providers to increase computing capacity according to demand.

Conclusion :

Data centers are the physical foundation of cloud computing because they store, process & manage the resources needed to deliver cloud services.

5)

Model	Definition	Example
Iaas	Infrastructure as a service provides virtual servers, storage & networking	Amazon EC2
Paas	Platform as a service provides a platform for developing & deploying application	Google App Engine
Saas	Software as a service provides ready-to-use software through the internet	Google Docs

Xaas	Anything as a service refers to various IT services delivered over the Internet	Microsoft Azure services

Explanation:

→ Iaas: Users manage their operating systems & Application, while the provider manages physical hardware.

→ Paas: Developers Create & deploy application without managing the underlying infrastructure.

→ Saas: Users access Software online without installing or maintaining the underlying infrastructure

→ Xaas: A board term Covering different services delivered as needed, including infrastructure, Platforms & software.

Conclusion:

These service models provide different levels of control & convenience depending on the users requirements.

10/10/24